Package 'limestab'

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Type Package

Title Stable Local Interpretable Model-Agnostic Explanations (LIME, OptiLIME, BayLIME, ALIME)

Version 0.1.0

Description Provides four local explanation methods for tabular data: LIME, OptiLIME, BayLIME, and ALIME, together with the stability metrics used to compare them across repeated explanations of the same instance: the Coefficient Stability Index (CSI), the Variable Stability Index (VSI), Median Rank Matching Stability (MRMS), Point-Based Stability Index (PCSI), and Interval Coefficient Stability Index (ICSI).
The methodology accompanies Toparlak and Purutcuoglu, ``A Median-Based Framework for Assessing the Stability in XAI Tools" (manuscript under revision); the citation will be updated upon publication.

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`limestab-package` *limestab: Stable LIME, OptiLIME, BayLIME and ALIME*

Description

Tools for local explanations of tabular black-box models using LIME, OptiLIME, BayLIME and ALIME, together with stability diagnostics for repeated explanations.

Details

The package provides coefficient, variable-selection, ranking, point-based and interval-based stability measures for repeated local explanations.

`alime_explainer` *Create an ALIME explainer*

Description

Create an ALIME explainer.

Usage

```
alime_explainer(training_data,
  feature_names = NULL,
  n_neighbors = 300,
  feature_selection = "forward_selection",
  penalty = 1,
  standardize = TRUE,
  embed_fn = NULL,
  random_state = NULL)
```

Arguments

`training_data`	Numeric training-data matrix.
`feature_names`	Optional character vector of feature names.
`n_neighbors`	Number of nearest perturbed points kept for the local model.
`feature_selection`	Feature-selection method.
`penalty`	Ridge penalty of the local model.
`standardize`	If TRUE, measure ALIME neighbourhood distances in the standardised feature space.
`embed_fn`	Optional embedding function applied before measuring distances.
`random_state`	Optional default seed.

Value

An object of class alime_explainer.

alime_explain_instance *Explain a single instance with ALIME*

Description

Explain a single instance with ALIME.

Usage

```
alime_explain_instance(explainer,
  data_row,
  predict_fn,
  label = 2,
  num_features = 5,
  num_samples = 10000,
  seed = NULL)
```

Arguments

explainer	An alime_explainer object.
data_row	The instance to explain.
predict_fn	Prediction function.
label	Column of predict_fn's output to explain.
num_features	Number of features kept in the local model.
num_samples	Number of points perturbed before keeping the nearest n_neighbors.
seed	Optional seed for reproducibility.

Value

A list with the same structure as lime_explain_instance().

bayesian_optimisation *Bayesian optimisation of a scalar objective*

Description

Bayesian optimisation of a scalar objective.

Usage

```
bayesian_optimisation(n_iters,
  sample_loss,
  bounds,
  n_pre_samples = 5,
  random_state = 42,
  verbose = FALSE)
```

Arguments

n_iters	Number of Bayesian-optimisation iterations after the initial design.
sample_loss	Function of one scalar returning the value to maximise.
bounds	c(lower, upper) search bounds.
n_pre_samples	Number of random initial evaluations.
random_state	Seed for the whole procedure.
verbose	If TRUE, print progress.

Value

A list with x_sample and y_sample, every point evaluated.

---

bayesian_ridge_fit *Bayesian ridge regression (Tipping 2001 / MacKay 1992)*

Description

Bayesian ridge regression (Tipping 2001 / MacKay 1992).

Usage

```
bayesian_ridge_fit(x,
  y,
  w,
  alpha_init = NULL,
  lambda_init = NULL,
  fit_alpha = TRUE,
  fit_lambda = TRUE,
  max_iter = 1000,
  tol = 1e-4,
  alpha_1 = 1e-6,
  alpha_2 = 1e-6,
  lambda_1 = 1e-6,
  lambda_2 = 1e-6,
  weighted_centering = FALSE)
```

Arguments

x	Numeric predictor matrix.
y	Numeric response vector.
w	Sample weights.
alpha_init	Initial (or fixed) noise precision; NULL uses 1 / var(y).
lambda_init	Initial (or fixed) weight precision; NULL uses 1.
fit_alpha	Whether the noise precision is re-estimated from data.
fit_lambda	Whether the weight precision is re-estimated from data.
max_iter	Maximum number of update iterations.
tol	Convergence tolerance on the coefficients.
alpha_1	Gamma-prior shape hyperparameter for alpha.
alpha_2	Gamma-prior rate hyperparameter for alpha.
lambda_1	Gamma-prior shape hyperparameter for lambda.
lambda_2	Gamma-prior rate hyperparameter for lambda.
weighted_centering	If TRUE, centre x/y with weighted means; if FALSE, use unweighted means.

Value

A list with intercept, coef, sigma (posterior covariance), alpha and lambda.

---

baylime_explainer *Create a BayLIME explainer*

Description

Create a BayLIME explainer.

Usage

```
baylime_explainer(training_data,
  feature_names = NULL,
  kernel_width = NULL,
  feature_selection = "auto",
  sample_around_instance = FALSE,
  model_regressor = c(
    "Bay_info_prior",
    "non_Bay",
    "Bay_non_info_prior",
    "BayesianRidge_inf_prior_fit_alpha"
  ),
  prior_alpha = 1,
  prior_lambda = 1000,
  random_state = NULL)
```

Arguments

training_data	Numeric training-data matrix.
feature_names	Optional character vector of feature names.
kernel_width	Optional kernel width.
feature_selection	Feature-selection method.
sample_around_instance	If TRUE, perturbations are centred on the instance.
model_regressor	One of "Bay_info_prior", "non_Bay", "Bay_non_info_prior" or "BayesianRidge_inf_prior_fit_alpha".
prior_alpha	Informative prior on the noise precision (used by "Bay_info_prior").
prior_lambda	Informative prior on the weight precision (used by "Bay_info_prior" and "BayesianRidge_inf_prior_fit_alpha").
random_state	Optional default seed.

Value

An object of class baylime_explainer.

---

baylime_explain_instance

Explain a single instance with BayLIME

Description

Explain a single instance with BayLIME.

Usage

```
baylime_explain_instance(explainer,
  data_row,
  predict_fn,
  label = 2,
  num_features = 5,
  num_samples = 5000,
  seed = NULL)
```

Arguments

explainer	A baylime_explainer object.
data_row	The instance to explain.
predict_fn	Prediction function.
label	Column of predict_fn's output to explain.
num_features	Number of features kept in the local model.
num_samples	Number of perturbed samples.
seed	Optional seed for reproducibility.

Value

A list with the same structure as lime_explain_instance(), plus local_std, coef_var, posterior_alpha and posterior_lambda.

`calculate_distances` *Euclidean distance of every row to an instance*

Description

Euclidean distance of every row to an instance.

Usage

```
calculate_distances(scaled_data,
  instance_scaled)
```

Arguments

`scaled_data`	Numeric matrix in standardised space.
`instance_scaled`	Standardised instance to measure distances from.

Value

A numeric vector of Euclidean distances.

`compute_cov_metrics` *Coefficient-of-variation metrics across repeated runs*

Description

Coefficient-of-variation metrics across repeated runs.

Usage

```
compute_cov_metrics(run_rankings,
  topk = NULL,
  eps = 1e-12,
  missing_value = 0)
```

Arguments

`run_rankings`	List of data frames (feature, beta), one per run.
`topk`	If not NULL, keep only the topk features with the largest median abs(beta).
`eps`	Small constant used to avoid division by zero.
`missing_value`	Value used for a feature absent from a given run.

Value

A list with cov_results (per feature) and the averages cov_unscaled, cov_scaled, cov_iqr, median_sd.

`compute_csi` *Coefficient Stability Index (CSI)*

Description

Coefficient Stability Index (CSI).

Usage

```
compute_csi(confidence_intervals,
  index_verbose = FALSE)
```

Arguments

`confidence_intervals`	List of named confidence-interval lists, one element per repeated LIME run.
`index_verbose`	If TRUE, print the per-feature overlap percentage.

Value

A single numeric value between 0 and 100.

`compute_icsi` *Interval-Based Coefficient Stability Index (ICSI)*

Description

Interval-Based Coefficient Stability Index (ICSI).

Usage

```
compute_icsi(B,
  train_ratio = 0.8,
  eps = 1e-12)
```

Arguments

`B`	Coefficient matrix (runs x features).
`train_ratio`	Share of runs used to build the reference interval.
`eps`	Small constant used to avoid division by zero.

Value

A named list of coverage results (ICSI_MAD, ICSI_rMAD, ICSI_IQR, ICSI_sMAD), each with coverage_per_feature and overall_coverage.

---

compute_mad_stdevs *Robust standard deviation from repeated coefficients*

Description

Robust standard deviation from repeated coefficients.

Usage

```
compute_mad_stdevs(beta_list)
```

Arguments

beta_list	Matrix of coefficients, one row per repeated explanation.

Value

A numeric vector, 1.4826 * MAD for each column.

---

compute_mrms *Median Rank Matching Stability of features across repeated runs*

Description

Median Rank Matching Stability of features across repeated runs.

Usage

```
compute_mrms(run_rankings,
  k = 5)
```

Arguments

run_rankings	List of data frames (feature, beta), one per run.
k	Number of top features used for the explanation-based top-k score.

Details

Features are ranked by decreasing absolute coefficient. Tied absolute coefficients receive the average of the rank positions they occupy. A missing feature is not assigned an artificial rank; it contributes zero to the matching score for that run. MRMS-All averages the resulting binary matches over all features and runs.

Value

A list with per_feature, mrms_all and mrms_topk. Scores are percentages.

`compute_pcsi` *PCSI stability indices*

Description

PCSI stability indices.

Usage

```
compute_pcsi(cov)
```

Arguments

`cov`	The list returned by compute_cov_metrics().

Value

A list with PCSI_MAD, PCSI_sMAD, PCSI_IQR and PCSI_MSD.

`compute_vsi` *Variable Stability Index (VSI)*

Description

Variable Stability Index (VSI).

Usage

```
compute_vsi(confidence_intervals,
  index_verbose = FALSE)
```

Arguments

`confidence_intervals`	List of named confidence-interval lists, one element per repeated LIME run.
`index_verbose`	If TRUE, print the overall percentage.

Details

For a pair of runs with different numbers of selected variables, the overlap is normalised by the smaller selected set, avoiding dependence on the first run's feature count.

Value

A single numeric value between 0 and 100.

compute_wls_stdevs *Standard deviations of weighted ridge coefficients*

Description

Standard deviations of weighted ridge coefficients.

Usage

```
compute_wls_stdevs(x,
  y,
  weights,
  alpha)
```

Arguments

x	Numeric predictor matrix used by the local model.
y	Numeric response vector.
weights	Sample weights.
alpha	Ridge penalty used when fitting the local model.

Value

A numeric vector of approximate coefficient standard deviations.

default_kernel_width *Default LIME kernel width for tabular data*

Description

Default LIME kernel width for tabular data.

Usage

```
default_kernel_width(n_features)
```

Arguments

n_features	Number of features in the data set.

Value

A single numeric value, 0.75 * sqrt(n_features).

fit_gp *Fit a 1-D Gaussian process (Matern 3/2 kernel)*

Description

Fit a 1-D Gaussian process (Matern 3/2 kernel).

Usage

```
fit_gp(x,
  y,
  alpha = 1e-6)
```

Arguments

x	Numeric vector of training inputs (kernel widths already evaluated).
y	Numeric vector of observed values (the loss at each x).
alpha	Observation noise variance added to the kernel diagonal.

Value

An object of class optilime_gp.

fit_weighted_ridge *Weighted ridge regression with an unpenalised intercept*

Description

Weighted ridge regression with an unpenalised intercept.

Usage

```
fit_weighted_ridge(x,
  y,
  w,
  alpha = 0)
```

Arguments

x	Numeric predictor matrix.
y	Numeric response vector.
w	Numeric vector of sample weights.
alpha	Ridge penalty; alpha = 0 is weighted least squares.

Value

A list with intercept, coef and fitted (fitted values).

`generate_perturbations` *Generate a perturbed neighbourhood in standardised feature space*

Description

Generate a perturbed neighbourhood in standardised feature space.

Usage

```
generate_perturbations(instance_scaled,
  num_samples = 5000,
  sample_around_instance = TRUE)
```

Arguments

`instance_scaled`	Standardised instance (numeric vector).
`num_samples`	Number of perturbed samples to generate; row 1 is the instance itself.
`sample_around_instance`	If TRUE, sample around the instance; otherwise around the origin (the training-data mean).

Value

A numeric matrix (num_samples x length(instance_scaled)) in standardised space.

`lime_check_stability` *Repeat an explanation and compute every stability metric*

Description

Repeat an explanation and compute every stability metric.

Usage

```
lime_check_stability(explainer,
  data_row,
  predict_fn,
  label = 2,
  num_features = 5,
  num_samples = 5000,
  n_calls = 10,
  index_verbose = FALSE,
  topk = 5,
```

```
  train_ratio = 0.8,
  seed = explainer$random_state,
  explain_fn = lime_explain_instance)
```

Arguments

explainer	An explainer object (from lime_explainer(), baylime_explainer() or alime_explainer()).
data_row	The instance to explain.
predict_fn	Prediction function passed on to explain_fn.
label	Column of predict_fn's output to explain.
num_features	Number of features kept in each individual explanation.
num_samples	Number of perturbed samples per explanation.
n_calls	Number of repeated explanations.
index_verbose	Passed on to compute_csi() / compute_vsi().
topk	Number of top features used for the top-k metrics.
train_ratio	Share of runs used as the reference set for compute_icsi().
seed	Seed for the whole procedure; the global RNG state is restored afterwards.
explain_fn	Function used to explain one instance, e.g. lime_explain_instance, baylime_explain_instance or alime_explain_instance.

Value

A list with csi, vsi, mrms_all, mrms_topk, PCSI_MAD/PCSI_sMAD/PCSI_IQR/PCSI_MSD, interval_results (ICSI) and the raw run_rankings / confidence_intervals.

---

lime_confidence_intervals

Confidence intervals for a LIME-style explanation

Description

Confidence intervals for a LIME-style explanation.

Usage

```
lime_confidence_intervals(explanation)
```

Arguments

explanation	The list returned by lime_explain_instance() or baylime_explain_instance().

Details

For Bayesian BayLIME fits, intervals are built from the posterior coefficient covariance. For LIME, ALIME and non-Bayesian BayLIME fits, approximate weighted-ridge standard errors are used.

Value

A named list of c(lower, upper) intervals, or NULL if uncertainty cannot be estimated.

`lime_explainer` *Create a LIME explainer*

Description

Create a LIME explainer.

Usage

```
lime_explainer(training_data,
  feature_names = NULL,
  kernel_width = NULL,
  feature_selection = "auto",
  sample_around_instance = FALSE,
  penalty = 1,
  random_state = NULL)
```

Arguments

`training_data`	Numeric training-data matrix used to centre and scale features.
`feature_names`	Optional character vector of feature names.
`kernel_width`	Optional kernel width; defaults to default_kernel_width().
`feature_selection`	Feature-selection method passed to select_features().
`sample_around_instance`	If TRUE, perturbations are centred on the instance.
`penalty`	Ridge penalty of the local model (0 = weighted least squares).
`random_state`	Optional default seed used by lime_check_stability().

Value

An object of class lime_explainer.

lime_explain_instance *Explain a single instance with LIME*

Description

Explain a single instance with LIME.

Usage

```
lime_explain_instance(explainer,
  data_row,
  predict_fn,
  label = 2,
  num_features = 5,
  num_samples = 5000,
  seed = NULL)
```

Arguments

explainer	A lime_explainer object created by lime_explainer().
data_row	The instance to explain, in the original feature space.
predict_fn	Prediction function; takes a matrix and returns a numeric vector or a matrix of class probabilities.
label	Column of predict_fn's output to explain, when it returns a matrix.
num_features	Number of features to keep in the local model.
num_samples	Number of perturbed samples to generate.
seed	Optional seed; if NULL, the global RNG stream continues (repeated calls give different neighbourhoods).

Value

A list describing the local explanation (coefficients, local_exp, score, local_pred, and the data needed for confidence intervals).

lime_kernel *LIME exponential kernel*

Description

LIME exponential kernel.

Usage

```
lime_kernel(distances,
  kernel_width)
```

Arguments

distances	Numeric vector of distances to the instance being explained.
kernel_width	Positive numeric kernel width.

Value

Numeric vector of proximity weights, sqrt(exp(-distances^2 / kernel_width^2)).

---

optilime *OptiLIME: choose the LIME kernel width by Bayesian optimisation*

Description

OptiLIME: choose the LIME kernel width by Bayesian optimisation.

Usage

```
optilime(explainer,
  data_row,
  predict_fn,
  bounds = c(0.1, 3.0),
  n_iters = 30,
  n_pre_samples = 10,
  label = 2,
  num_features = 5,
  num_samples = 5000,
  max_r2 = 0.9,
  epsilon = NULL,
  seed = 42,
  verbose = FALSE)
```

Arguments

explainer	A lime_explainer object.
data_row	The instance to explain.
predict_fn	Prediction function.
bounds	c(lower, upper) search bounds for the kernel width.
n_iters	Number of Bayesian-optimisation iterations.
n_pre_samples	Number of random initial kernel widths.
label	Column of predict_fn's output to explain.
num_features	Number of features kept in the local model.
num_samples	Number of perturbed samples per explanation.
max_r2	Target local R-squared passed to optilime_score().

epsilon	Optional tolerance passed to optilime_score().
seed	Seed used for every evaluation, so kernel widths are compared on identical perturbations.
verbose	If TRUE, print progress.

Value

A list with best_width, best_loss, r2, explanation, explainer and optimization.

`optilime_score` *OptiLIME objective (reflected local R-squared)*

Description

OptiLIME objective (reflected local R-squared).

Usage

```
optilime_score(explanation,
  max_r2 = 0.9,
  epsilon = NULL)
```

Arguments

explanation	The list returned by lime_explain_instance().
max_r2	Target local R-squared; scores above it are penalised symmetrically.
epsilon	Optional tolerance used to flag and penalise degenerate explanations.

Value

A single numeric value to be maximised by optilime().

`predict.optilime_gp` *Predict method for optilime_gp objects*

Description

Predict method for optilime_gp objects.

Usage

```
predict.optilime_gp(object,
  newdata,
  return_std = FALSE,
  ...)
```

Arguments

object	An optilime_gp object returned by fit_gp().
newdata	Numeric vector of new points.
return_std	If TRUE, also return the predictive standard deviation.
...	Unused; included for S3 consistency.

Value

A numeric vector of means, or a list with mean and sd when return_std = TRUE.

---

select_features *Select the features used by a LIME-style local model*

Description

Select the features used by a LIME-style local model.

Usage

```
select_features(x,
  y,
  w,
  num_features,
  method = "auto")
```

Arguments

x	Numeric predictor matrix.
y	Numeric response vector.
w	Sample weights.
num_features	Number of features to select.
method	One of "auto", "forward_selection", "highest_weights" or "none".

Value

An integer vector of selected column indices of x.

`weighted_r2` *Weighted R-squared*

Description

Weighted R-squared.

Usage

```
weighted_r2(y,
  fitted,
  w)
```

Arguments

`y`	Observed response.
`fitted`	Fitted values from the local model.
`w`	Sample weights.

Value

A single numeric value between (typically) 0 and 1.

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